

CALCULUS

Calculus:

The branch of mathematics in which we study derivative of a function and its related ideas is called calculus.

Real Line:

To every real number there will correspond a point on the line and conversely, every point on the line will represent a real number. Thus, there is one-to-one correspondence between the set of real numbers and the points of the straight line. This line is called real line.

**Set:**

A well defined collection of distinct objects is called a set.

Cartesian Product of sets:

The Cartesian product of sets is a set of ordered pairs from a first set to second set. That is $A \times B = \{(a, b); a \in A \text{ and } b \in B\}$.

Relation:

Any subset of $A \times B$ is called a relation from A to B.

Function:

A rule $f: A \rightarrow B$ is called a function from A to B if

- (i) $f \subseteq A \times B$
- (ii) Domain of $f = A$
- (iii) If $(x, y_1), (x, y_2) \in f \Leftrightarrow y_1 = y_2$.

Onto Function (Surjective):

A function $f: A \rightarrow B$ is called onto if Range of $f = B$.

Into Function:

If a function is not onto function.

One-to-one (Injective) function:

A function $f: A \rightarrow B$ is called into if $(x_1, y), (x_2, y) \in f \Leftrightarrow x_1 = x_2$.

Bijjective Function:

A function which is onto one-to-one is called bijjective function.

The following table shows the information of some counting facts related to functions.

Number of elements in set A	n
Number of elements in set B	m
Number of subsets of A	2^n
Number of subset of B	2^m
Number of elements in $A \times B$	nm

Number of elements in $B \times A$	mn
Number of relations from A to B	2^{nm}
Number of relations from B to A	2^{mn}
Number of functions from A to B	m^n
Number of functions from B to A	n^m
Number of one-to-one functions from A to B	${}^mP_n = \frac{m!}{(m-n)!}$ where $m \geq n$
Number of onto functions from A to B	$\sum_{r=1}^m (-1)^{m-r} {}^mC_r r^n$ where $1 \leq m \leq n$
Number of into functions from A to B	$m^n - (\text{no. of onto function})$
Number of bijective functions from A to B	If $n = m$, then $n!$

Sequence:

A sequence is a function where domain is a subset of natural number or whole number.

Limit of function:

The function f is said to have limit l as x approaches a if for every $\epsilon > 0$, there exists a positive real number δ such that $|f(x) - l| < \epsilon$ whenever $0 < |x - a| < \delta$.

Following table shows the information of some useful limits.

$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$
$\lim_{n \rightarrow 0} (1 + n)^{1/n} = e$
$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a, a > 0, a \neq 1$
$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$
$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
$\lim_{x \rightarrow 0} \frac{x^p - a^p}{x - a} = pa^{p-1}$

Continuity:

A function f is said to be continuous at point $a \in \text{Dom } f$ if

- (i) the point a lies in an open interval contained in $\text{dom } f$

$$(ii) \quad \lim_{n \rightarrow a} f(x) = f(a).$$

If a function is continuous at all points of its domain, then f is called continuous function.

Following table shows the information of some continuous functions in R .

Constant function	$f(x) = k, k \in R$
Identity function	$f(x) = x$
Polynomial functions	e. $g f(x) = 4x^4 + 3x^3 + 2x^2 + x$
Trigonometric functions	$f(x) = \sin x$ and $f(x) = \cos x$

Derivative:

A function $f(x)$ is called differential (or a function has a derivative) if

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Provided the limit exists.

Following table shows the information of general theorems of derivatives.

$\frac{d}{dx} (x^p) = px^{p-1}$	, $p \in R$
$\frac{d}{dx} (c) = 0$	, $c \in R$
$\frac{d}{dx} (u(x) v(x)) = u(x) v'(x) + v(x) u'(x)$	
$\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{v(x) u'(x) - u(x) v'(x)}{(v(x))^2}$	
$\frac{d}{dx} (f \circ g(x)) = [f'(g(x))] g'(x)$	
$\frac{d}{dx} (\log_a f(x)) = \frac{f'(x)}{f(x)} \log_a e$	
$\frac{d}{dx} (a^{f(x)}) = a^{f(x)} \log_e a f'(x)$	

Higher Derivative:

Let $y = f(x)$ be differentiable in some interval $[a, b]$. The $\frac{dy}{dx} = f'(x)$ may also possess derivative in $[a, b]$. If we apply the definition of derivative to $f'(x)$, the resulting limit (if it exists) is called second derivative.

$$\text{i.e. } \frac{d^2y}{dx^2} = f''(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h}$$

Continuing in this way, we can evaluate further derivative.

Following table shows the information of some standard. nth derivatives.

$\frac{d^n}{dx^n} (ax + b)^m = \frac{m!}{(m-n)!} a^n (ax + b)^{m-n}, m \geq n$
$\frac{d^n}{dx^n} (\ln(ax + b)) = \frac{(-1)^{n-1} (n-1)! a^n}{(ax + b)^n}$
$\frac{d^n}{dx^n} (e^{ax}) = a^n e^{ax}$
$\frac{d^n}{dx^n} (\sin(ax + b)) = a^n \sin(ax + b + n \cdot \frac{\pi}{2})$
$\frac{d^n}{dx^n} (\cos(ax + b)) = a^n \cos(ax + b + n \cdot \frac{\pi}{2})$

Leibniz Rule:

Let $y = uv$ where u, v are functions of x and both possess derivatives of n th orders. Then

$$y^{(n)} = {}^nC_0 u^{(n)} v + {}^nC_1 u^{(n-1)} v' + \dots + {}^nC_{n-1} u' v^{(n-1)} + {}^nC_n uv^{(n)}.$$

Rolle's Theorem:

Let a function f be

- (i) Continuous on the closed interval $[a, b]$
- (ii) differentiable on open interval $]a, b[$
- (iii) $f(a) = f(b)$

Then there exists at least one point $c \in]a, b[$ such that $f'(c) = 0$.

The Mean Value Theorem:

Let a function f be

- (i) Continuous on $[a, b]$
- (ii) differentiable on $]a, b[$ then there exists a point $c \in]a, b[$ such that $\frac{f(b) - f(a)}{b - a} = f'(c)$.

Cauchy's Mean Value Theorem:

If two functions f and g are

- (i) Continuous on $[a, b]$
- (ii) differential on $]a, b[$
- (iii) $g'(x) \neq 0$ for all $x \in]a, b[$ then there exists at least one point $C \in]a, b[$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Taylor's Theorem:

If a function f is such that

- (i) $f, f', f'', \dots, f^{(n-1)}$ are continuous on $[a, a+h]$
 (ii) $f^{(n)}$ exists on $]a, a+h[$ then there exists a number $\theta, 0 < \theta < 1$, such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} f^{(n)}(a + \theta h).$$

Antiderivatives:

Let $f(x)$ be given function. If there is a differentiable function $F(x)$ such that $\frac{d}{dx}[F(x)] = f(x)$ then $F(x)$ is called an anti derivative of $f(x)$. And the expression $f(x) + c$ is called e integral of $f(x)$ and is called indefinite integral of $f(x)$ and is denoted by $\int f(x) dx = F(x) + c$

Table of Integral:

Following is the information of some standard integrals.

$$1. \int u^n du = \frac{1}{n+1} u^{n+1} + C, \text{ where } n \neq -1$$

$$2. \int u^{-1} du = \int \frac{1}{u} du = \ln |u| + C$$

$$3. \int e^u du = e^u + C$$

$$4. \int a^u du = \frac{1}{\ln a} a^u + C, \text{ where } a > 0 \text{ and } a \neq 1$$

$$5. \int \sin u du = -\cos u + C$$

$$6. \int \cos u du = \sin u + C$$

$$7. \int \sec^2 u du = \tan u + C$$

$$8. \int \sec u \tan u du = \sec u + C$$

$$9. \int \csc^2 u du = -\cot u + C$$

$$10. \int \csc u \cot u \, du = -\csc u + C$$

$$11. \int f(au + b) \, du = \frac{1}{a} F(au + b) + C, \text{ where } F(u) \text{ is an antiderivative of } f(u)$$

$$12. \int \tan u \, du = \ln |\sec u| + C = -\ln |\cos u| + C$$

$$13. \int \sec u \, du = \ln |\sec u + \tan u| + C$$

$$14. \int \cot u \, du = -\ln |\csc u| + C = \ln |\sin u| + C$$

$$15. \int \csc u \, du = \ln |\csc u - \cot u| + C$$

$$16. \int \frac{1}{\sqrt{1-u^2}} \, du = \sin^{-1} u + C$$

$$17. \int \frac{1}{\sqrt{a^2-u^2}} \, du = \sin^{-1} \left(\frac{u}{a} \right) + C \quad \text{for } a > 0$$

$$18. \int \frac{1}{1+u^2} \, du = \tan^{-1} u + C$$

$$19. \int \frac{1}{a^2+u^2} \, du = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C \quad \text{for } a > 0$$

$$20. \int \frac{1}{u\sqrt{u^2-1}} \, du = \sec^{-1} u + C \quad \text{for } u > 1$$

$$21. \int \frac{1}{u\sqrt{u^2-a^2}} \, du = \frac{1}{a} \sec^{-1} \left(\frac{u}{a} \right) + C \quad \text{for } u > a > 0$$

Basic Rational Forms

$$22. \int \frac{1}{a^2-u^2} \, du = \frac{1}{2a} \ln \left| \frac{a+u}{a-u} \right| + C \quad \text{for } a > 0$$

$$23. \int \frac{1}{u^2 - a^2} du = \frac{1}{2a} \ln \left| \frac{u - a}{u + a} \right| + C \quad \text{for } a > 0$$

Trigonometric Forms

$$24. \int \sin^2 u \, du = \frac{1}{2}u - \frac{1}{4} \sin 2u + C$$

$$25. \int \cos^2 u \, du = \frac{1}{2}u + \frac{1}{4} \sin 2u + C$$

$$26. \int \sec^3 u \, du = \frac{1}{2} \sec u \tan u + \frac{1}{2} \ln |\sec u + \tan u| + C$$

Inverse Trigonometric Forms

$$27. \int \sin^{-1} u \, du = u \sin^{-1} u + \sqrt{1 - u^2} + C$$

$$28. \int \cos^{-1} u \, du = u \cos^{-1} u - \sqrt{1 - u^2} + C$$

$$29. \int \tan^{-1} u \, du = u \tan^{-1} u - \frac{1}{2} \ln (u^2 + 1) + C$$

$$30. \int \cot^{-1} u \, du = u \cot^{-1} u + \frac{1}{2} \ln (u^2 + 1) + C$$

$$31. \int \sec^{-1} u \, du = u \sec^{-1} u - \ln (u + \sqrt{u^2 - 1}) + C \quad \text{for } u > 1$$

$$32. \int \csc^{-1} u \, du = u \csc^{-1} u + \ln (u + \sqrt{u^2 - 1}) + C \quad \text{for } u > 1$$

Forms Involving $\sqrt{a^2 - u^2}$

$$33. \int \sqrt{a^2 - u^2} \, du = \frac{1}{2}u\sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{u}{a} \right) + C \quad \text{for } a > 0$$

Forms Involving $\sqrt{a^2 + u^2}$

$$34. \int \sqrt{a^2 + u^2} \, du = \frac{1}{2} u \sqrt{a^2 + u^2} + \frac{a^2}{2} \ln \left(u + \sqrt{a^2 + u^2} \right) + C$$

Forms Involving $\sqrt{u^2 - a^2}$

$$35. \int \sqrt{u^2 - a^2} \, du = \frac{1}{2} u \sqrt{u^2 - a^2} - \frac{a^2}{2} \ln \left(u + \sqrt{u^2 - a^2} \right) + C \quad \text{for } u > a > 0$$

Forms Involving $au + b$

$$36. \int \sqrt{au + b} \, du = \frac{2}{3a} (au + b)^{3/2} + C$$

$$37. \int u \sqrt{au + b} \, du = \frac{2}{15a^2} (3au - 2b) (au + b)^{3/2} + C$$

Exponential Forms

$$38. \int e^{au} \, du = \frac{1}{a} e^{au} + C$$

$$39. \int u e^{au} \, du = \frac{1}{a^2} (au - 1) e^{au} + C$$

$$40. \int u^2 e^{au} \, du = \frac{1}{a^3} (a^2 u^2 - 2au + 2) e^{au} + C$$

$$41. \int e^{au} \sin bu \, du = \frac{1}{a^2 + b^2} e^{au} (a \sin bu - b \cos bu) + C$$

$$42. \int e^{au} \cos bu \, du = \frac{1}{a^2 + b^2} e^{au} (a \cos bu + b \sin bu) + C$$

Logarithmic Forms

$$43. \int \ln u \, du = u \ln u - u + C$$

Definite Integrals

$$44. \int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$$

Properties of Definite Integrals:

Certain properties are useful in solving problems requiring the application of the definite integral. Some of the more common properties are

$$1. \int_a^a f(x) \, dx = 0$$

$$2. \int_a^b f(x) \, dx = - \int_b^a f(x) \, dx$$

$$3. \int_a^b c \, dx = c(b-a), \text{ where } c \text{ is a constant}$$

$$4. \int_a^b cf(x) \, dx = c \int_a^b f(x) \, dx$$

$$5. \text{Sum Rule: } \int_a^b [f(x) + g(x)] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$$

$$6. \text{Difference Rule: } \int_a^b [f(x) - g(x)] \, dx = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx$$

$$7. \text{If } f(x) \geq 0 \text{ on } [a, b], \text{ then } \int_a^b f(x) \, dx \geq 0$$

$$8. \text{If } f(x) \leq 0 \text{ on } [a, b], \text{ then } \int_a^b f(x) \, dx \leq 0$$

9. If $f(x) \geq g(x)$ on $[a, b]$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$

10. If a , b , and c are any three points on a closed interval, then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Conic Sections:

The circle, parabola, ellipse and hyperbola are called conic sections because each can be obtained as the curve of intersection of a cone and a plane. A cone is generated by all lines through a fixed point and the circumference of a circle.

Circle:

A set of all points in the planes which are equidistant from a fixed point is called circle.

Parabola:

A set of all points in the plane which equidistant from a fixed point and fixed line is called parabola.

Ellipse:

A set of all points in the plane whose sum of distances from two fixed points is constant.

Hyperbola:

A set of all points in the plane whose difference of distances from two fixed points is constant.

Standard equations of some curves.

Auxiliary Circle:

The circle described on the major axis of an ellipse is called the auxiliary circle.

Cylinder:

Let C be a curve in a place and let L be a line not in the plane. The union of all lines that intersect C and are parallel to L is called a cylinder.

Cone:

A surface generated by a straight line passing through a fixed point and intersecting a fixed plane curve not containing the fixed point is called a cone.

Sphere:

The set of a all points in space that are equidistant from a fixed point is called a sphere.

Following table shows the information of equations of some standard curves.

Curve	Equation
Conic section	$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
Circle	$x^2 + y^2 + 2gx + 2fy + c = 0$ or $(x - h)^2 + (y - k)^2 = r^2$

Parabola	$y^2 = 4ax$ or $x^2 = 4ay$
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
Pair of straight lines	$ax^2 + 2hxy + by^2 = 0$
Cardioid	$r = a(p + q \sin \theta)$ where $a > 0$ and $p \geq 2q$
Lemniscate	$r^2 = a^2 \cos 2\theta$
Spiral	$r = a\theta, a > 0$ Or $a = r\theta, a > 0$ Or $r = ae^\theta, a > 0$
Three-Leafed Rose	$r = a \sin 3\theta, a > 0$ or $r = a \cos 3\theta, a > 0$
Four-Leafed Rose	$r = a \sin 2\theta, a > 0$ $r = a \cos 2\theta, a > 0$
Five-Leafed Rose	$r = a \sin 5\theta, a > 0$ $r = a \cos 5\theta, a > 0$
Line	$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = t$
Plane	$ax + by + cz = d$
Sphere	$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Elliptic Paraboloid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = cz^2$
Elliptic Cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = Cz^2$
Hyperboloid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
Hyperbolic Paraboloid	$Z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$

Multiple Choice Questions:

1. $\lim_{x \rightarrow 0} \frac{1}{x} =$, $x > 0$
(a) 0 (b) ∞
(c) $-\infty$ (d) 1
2. $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) =$
(a) 0 (b) ∞
(c) 1 (d) 2
3. $\lim_{x \rightarrow 3^-} \frac{3-x}{|3-x|} =$
(a) 0 (b) 1
(c) -1 (d) 2
4. $\lim_{h \rightarrow 0} \frac{|-1+h| - 1}{h} =$
(a) 0 (b) 1
(c) -1 (d) 2
5. $\lim_{x \rightarrow 1} [x] [x+1] =$ where $[.]$ is bracket function.
(a) -1 (b) 0
(c) 1 (d) 2
6. Which of the following is true:
(a) $(|f|)(x) = f|x|$ (b) $(|f|)(x) = f(x)$
(c) $(|f|)(x) = |f(x)|$ (d) none
7. If $[x]$ is a bracket function then $f(x) = x - [x]$ is discontinuous at:
(a) R (b) N
(c) Q (d) Z
8. Let $f(x) = x^2$ and $g(x) = \begin{cases} -4, & x \leq 0 \\ |x-4|, & x > 0 \end{cases}$ then which of the following is continuous at $x = 0$.
(a) $f(x)$ (b) $\text{fog}(x)$
(c) $\text{gof}(x)$ (d) all of these
9. The solution of inequality $|2x+5| > |2-5x|$ is:
(a) $[-\infty, 0]$ (b) $\left(\frac{-1}{2}, \frac{1}{2}\right)$
(c) $\left(\frac{-3}{7}, \frac{7}{3}\right)$ (d) $(-1, \infty)$
10. Which of the following satisfies the inequality $|x| + |x-1| > 1$
(a) $(-\infty, 0)$ (b) $(0, 1)$
(c) $(0, \infty)$ (d) none
11. Which of the following satisfies the inequality $|5x+6| \geq 5$.

- (a) $(-1, \infty)$ (b) $\left(\frac{-1}{2}, \frac{1}{5}\right)$
 (c) $\left(\frac{-1}{5}, \infty\right)$ (d) $(-6, \infty)$
12. Solution of inequality $\frac{x^2 - 2}{1 - 2x} > 1$ is:
 (a) $(-\infty, \infty)$ (b) $(-\infty, -3) \cup (0.5, 1)$
 (c) $(-\infty, -1) \cup (1, 5)$ (d) none
13. $f(x) = |x| + |x - 1|$ is not differential at:
 (a) -1 (b) 2
 (c) 1 (d) 4
14. The derivative of $(x)^{x^2}$ is:
 (a) $2(x)^{x^2}$ (b) $(x)^{(x^2 + 1)} [2 \ln x + 1]$
 (c) $(x)^{x^2} [\ln x + 2]$ (d) none
15. If $x < 0$, then derivative of $(x + |x|)^{1/2}$ is:
 (a) 1 (b) 0
 (c) $(2x)^{-1/2}$ (d) $\frac{1}{2}(x + |x|)^{3/2}$
16. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} =$
 (a) 1 (b) 0
 (c) 1 (d) x
17. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} =$
 (a) -1 (b) 0
 (c) 1 (d) does not exist
18. $\lim_{(x,y) \rightarrow (2,1)} \frac{x^3 + 2x^2y}{x + 2y} =$
 (a) -1 (b) 0
 (c) 2 (d) 4
19. $\lim_{x \rightarrow 0} x^x =$
 (a) -1 (b) 0
 (c) 1 (d) e
20. $\int \frac{dx}{x^2 - 1} =$
 (a) $\cos hx$ (b) $\frac{1}{2} \ln \left| \frac{x-1}{x+1} \right|$
 (c) $\sin h^{-1}x$ (d) none
21. $\int \ln x \, dx =$

- (a) $\frac{1}{x}$ (b) $\ln x$
(c) $x \ln x - x$ (d) $\ln(\ln x)$
22. The n th derivative of $(ax + b)^m$ is:
(a) $m(ax + b)^{m-n}a$ (b) $m!(ax + b)^{m-n}$
(c) $\frac{m!}{(m-n)!}a^n(ax + b)^{m-n}$ (d) none
23. The n th derivative of e^{ax} is:
(a) ae^{ax} (b) $a^n e^{ax}$
(c) $a^{n-1}e^{ax}$ (d) 0
24. The n th derivative of a polynomial of degree n is:
(a) 1 (b) 0
(c) constant (d) linear
25. If $y = f(x, y)$ then which of the following is correct.
(a) $\frac{dy}{dx} = f_x f_y$ (b) $\frac{dy}{dx} = f_{xy}$
(c) $\frac{dy}{dx} = -\frac{f_x}{f_y}$ (d) $\frac{dy}{dx} = \frac{f_y}{f_x}$
26. If $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ then $f_{xy}(0, 0) =$
(a) $-k$ (b) -1
(c) h (d) 1
27. If $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ then $f_y(h, 0) =$
(a) k (b) 0
(c) 1 (d) h
28. $\lim_{x \rightarrow 0} \log_{\tan x}(\tan 2x) =$
(a) -1 (b) 0
(c) 1 (d) 2
29. $\lim_{x \rightarrow 0} \left[\frac{a}{x} - \cot\left(\frac{x}{a}\right) \right] =$
(a) -2 (b) -1
(c) 0 (d) 1
30. $\lim_{x \rightarrow \pi/2} (\sec x)^{\cot x} =$
(a) -2 (b) -1
(c) 0 (d) 1
31. $\lim_{x \rightarrow \pi/2} (1 - \sin x)^{\cos x} =$
(a) -1 (b) 0
(c) 1 (d) ∞

32. $\lim_{x \rightarrow 0} x \ln (\tan x) =$
(a) -1 (b) 0
(c) 1 (d) ∞
33. $\lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\tan x} =$
(a) -1 (b) 0
(c) 1 (d) ∞
34. $\int \pi^{\sin x} dx =$
(a) $\cos x$ (b) $\pi^{\cos x}$
(c) $\pi^{\sin x}$ (d) $\frac{\pi^{\sin x}}{\ln \pi} \cos x$
35. $\int (2.71)^x dx =$
(a) $x (2.71)^{x-1}$ (b) $\frac{(2.71)^x}{\ln (2.71)}$
(c) $(2.71)^x$ (d) none
36. Which of the following is a useful substitution to handle integral involving $\sqrt{x^2 + a^2}$.
(a) put $x = a \tan \theta$ (b) put $x = a \sin \theta$
(c) put $x = a \sec \theta$ (d) put $x = a \cos \theta$
37. Which of the following is a useful substitution to handle integral involving $\sqrt{x^2 - a^2}$
(a) put $x = a \cos hz$ (b) put $x = a \sin hz$
(c) put $x = a \sec hz$ (d) put $x = a \tan hz$
38. Which of the following is a suitable substitution if integral is of the form $\int \frac{dx}{\text{linear} \sqrt{\text{linear}}}$
(a) $\text{linear} = z$ (b) $\text{linear} = z + 1$
(c) $\sqrt{\text{linear}} = z$ (d) $\sqrt{\text{linear}} = z^2$
39. Which of the following is a suitable substitution if integral is of the form $\int \frac{dx}{\text{quadratic} \sqrt{\text{linear}}}$
(a) $\text{linear} = z$ (b) $\text{quadratic} = z$
(c) $\sqrt{\text{linear}} = z$ (d) $\text{quadratic} = z^2$
40. Which of the following is a suitable substitution if integral is of the form $\int \frac{dx}{\text{quadratic} \sqrt{\text{quadratic}}}$
(a) $\text{quadratic} = z$ (b) $x = \frac{1}{z}$
(c) $\text{quadratic} = \frac{1}{z}$ (d) $\sqrt{\text{quadratic}} = \frac{1}{z}$

41. Which of the following is a suitable substitution if integral is of the form $\int \frac{dx}{\text{linear} \sqrt{\text{quadratic}}}$

(a) $\text{linear} = \frac{-1}{z}$

(b) $\text{quadratic} = \frac{1}{z}$

(c) $x = \frac{1}{z}$

(d) none

42. Which of the following is a substitution for $\int \frac{dx}{\sqrt{x} - (x)^{1/3}}$

(a) $x = \frac{1}{z}$

(b) $x = z$

(c) $x = z^3$

(d) $x = z^6$

43. The substitution for $\int \frac{xdx}{(x^2 - 2x + 2) \sqrt{x-1}}$ is

(a) $x = z$

(b) $x - 1 = z$

(c) $x - 1 = z^2$

(d) none

44. The substitution for $\int \frac{dx}{x^2 \sqrt{x^2 + 1}}$ is

(a) $x = \frac{1}{z}$

(b) $x^2 = z$

(c) $x^2 + 1 = z$

(d) $x = z^2$

45. The substitution for $\int \frac{dx}{(x-1) \sqrt{x^2 + 1}}$

(a) $x = z$

(b) $x = \frac{1}{z}$

(c) $x^2 + 1 = z$

(d) $x = z^2$

46. The substitution for $\int \frac{dx}{1 + \sin x}$ is:

(a) $z = \tan x$

(b) $z = \tan\left(\frac{x}{2}\right)$

(c) $z = \sec x$

(d) $z = \sin x$

47. Which of the following is correct.

(a) $\int_a^b f(x) dx = - \int_b^a f(x) dx$

(b) $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$(c) \int_0^{2a} f(x) dx - \int_0^a f(x) dx = \int_0^a f(2a-x) dx$$

(d) all of these

48. If $f(2a-x) = f(x)$ then $\int_0^{2a} f(x) dx =$

(a) 0

(b) $2 \int_0^a f(x) dx$

(c) π

(d) none

49. If $f(x)$ is an odd function then $\int_{-a}^a f(x) dx =$

(a) $2 \int_0^a f(x) dx$

(b) 0

(c) $\int_0^a f(x) dx$

(d) none

50. $\int_1^{\infty} \frac{dx}{x^2} =$

(a) ∞

(b) 0

(c) 1

(d) none

51. $\int_{-\infty}^0 e^x dx =$

(a) $-\infty$

(b) 0

(c) 1

(d) ∞

52. Which of the following is a Gamma function (generalized factorial function).

(a) $\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$

(b) $\Gamma(x) = \int_0^{\infty} t^x e^{-t} dt$

(c) $\Gamma(x) = \int_{-\infty}^{\infty} t^{x+1} e^t dt$

(d) none

53. $\Gamma(1) =$

(a) 0

(b) 1

(c) 2!

(d) none

54. $\Gamma(3) =$

55. (a) 0 (b) 6
(c) 2 (d) 3!
56. $\int_0^{\pi/2} \sin^3 x \cos^4 x \, dx =$
(a) $n!$ (b) $(n+1)!$
(c) $(n-1)!$ (d) $\frac{n!}{2}$
57. $\int_0^{\pi/2} \sin^2 x \cos^3 x \, dx =$
(a) 0 (b) $\frac{1}{17}$
(c) $\frac{2}{35}$ (d) 12
58. $\Gamma\left(\frac{1}{2}\right) =$
(a) $\frac{2}{5}$ (b) $\frac{1}{5}$
(c) $\frac{7}{25}$ (d) $\frac{3}{14}$
59. $\Gamma\left(\frac{5}{2}\right) =$
(a) 0 (b) e
(c) $\sqrt{\pi}$ (d) 1
60. The general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents ellipse if:
(a) $\frac{3}{4}\sqrt{\pi}$ (b) $\frac{1}{2}\sqrt{\pi}$
(c) $\frac{5}{2}$ (d) π
61. Which of the following are parametric equations of parabola.
(a) $x = a \cos \theta, y = a \sin \theta$ (b) $x = a \sec \theta, y = a \tan \theta$
(c) $x = a \theta^2, y = 2a \theta$ (d) none
62. The equation $\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$ represents
(a) parabola (b) ellipse
(c) hyperbola (d) circle

63. $\frac{l}{r} = 1 - e \cos \theta$ is an equation of:
- (a) parabola (b) ellipse
(c) hyperbola (d) circle
64. $r = a(1 - \cos \theta)$ where $a > 0$ represents:
- (a) cardioid (b) four-leafed rose
(c) lemniscate of bernoulli (d) none
65. $r = 1 - 2 \cos \theta$ represents:
- (a) limaçon (b) five leafed rose
(c) cardioids (d) none
66. The equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ represents:
- (a) paraboloid (b) ellipsoid
(c) hyperboloid (d) none
67. The equation $z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$ represents:
- (a) paraboloid (b) hyperbolic paraboloid
(c) elliptic paraboloid (d) none
68. The formula $v = \int_a^b \pi [f(x)]^2 dx$ represents volume of a solid by:
- (a) disc method (b) washer method
(c) shell method (d) cross sectional method
69. The formula $v = \int_a^b 2\pi x f(x) dx$ represents volume of a solid by:
- (a) disc method (b) washer method
(c) shell method (d) cross sectional method
70. $\int_0^1 \int_1^2 dx dy =$
- (a) 0 (b) 1
(c) 9 (d) 10
71. $\int_0^1 \int_0^{x^2} xe^y dy dx =$
- (a) e (b) 1
(c) $\frac{e}{2} - 1$ (d) π

72. $\int_2^4 \int_y^{8-y} y \, dx \, dy =$

(a) 1

(b) 0

(c) $\frac{32}{3}$

(d) $\frac{64}{5}$

73. $\int_0^5 \int_x^{6x-x^2} dy \, dx =$

(a) $\frac{625}{6}$

(b) $\frac{125}{6}$

(c) $\frac{25}{6}$

(d) 0

74. $\int_0^1 \int_0^{2-2x} (x^2 + y^2 + 1) \, dy \, dx =$

(a) 0

(b) $\frac{7}{6}$

(c) $\frac{11}{6}$

(d) 24

75. $\int_{-1}^3 \int_2^4 \int_1^3 3(x^2y + y^2z) \, dz \, dy \, dx =$

(a) 0

(b) 12

(c) 24

(d) 42

76. $\int_{-4}^4 \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} \int_0^3 (x+y)z \, dz \, dy \, dx =$

(a) 0

(b) 10

(c) 9

(d) 32

KEY to MCQ of Calculus

1	b	2	c	3	c	4	c	5	b	6	c
7	d	8	d	9	c	10	a	11	c	12	b
13	c	14	b	15	b	16	b	17	d	18	d
19	c	20	b	21	c	22	?	23	b	24	c
25	c	26	b	27	?	28	c	29	c	30	d
31	c	32	b	33	c	34	d	35	c	36	?
37	?	38	c	39	c	40	?	41	c	42	d
43	c	44	a	45	b	46	b	47	d	48	b
49	b	50	c	51	c	52	a	53	b	54	c
55	c	56	c	57	a	58	c	59	a	60	b
61	c	62	c	63	b	64	a	65	a	66	b
67	b	68	a	69	c	70	b	71	c	72	c
73	b	74	c	75	c	76	a				